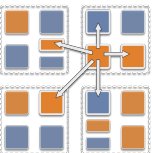


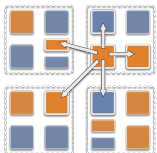
Profiling with Projections

- Projections: Java applet that visualizes Charm4Py profiling traces
 - Charm4Py programs consist of PEs that each process messages, e.g. entry method invocations
 - Beginning and end times of entry methods are tracked and output into a log file (1 per PE)
 - Information from these files are then processed by the Projections application



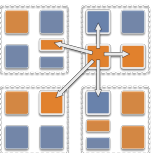
Files produced from profiling

- **.sts**: Overall information about program
 - Program name and command line
 - Charm version and machine layer
 - Command line parameters, including number of PEs
 - List of indices for each chare type and entry method
- **.projrc**: Global start and end time
- **.[log.gz](#)** (1 per PE): List of events, each with a timestep
 - Message creation
 - Message packing/unpacking
 - Beginning/end of message processing
 - Beginning/end of idle time
 - User-defined events



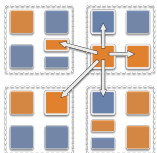
To enable projections in Charm4Py

- When building Charm++ backend, either build the non-production version, or include “--enable-tracing” in the build command
 - This will cause all Charm4Py runs to generate log files
- Projections runtime options
 - +traceroot: Directory which will contain trace files (recommended)
 - +logsize: Maximum allowed size of each log file in bytes (recommended for fine-grained applications)



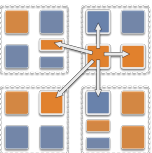
Projections install and launch

- Prerequisite: Java version 8+
- Download Projections binary from this link:
<https://charm.cs.illinois.edu/software>
- Launch Projections
 - From command line: add projections directory to PATH and run 'projections'
 - Or launch from desktop interface/finder window/etc.



Loading projections file

- Go to File > Open file(s)
- Navigate to directory with log files/traces
- Select .sts file
- Open “Tools” section in toolbar to select viewing mode



Example: Home screen

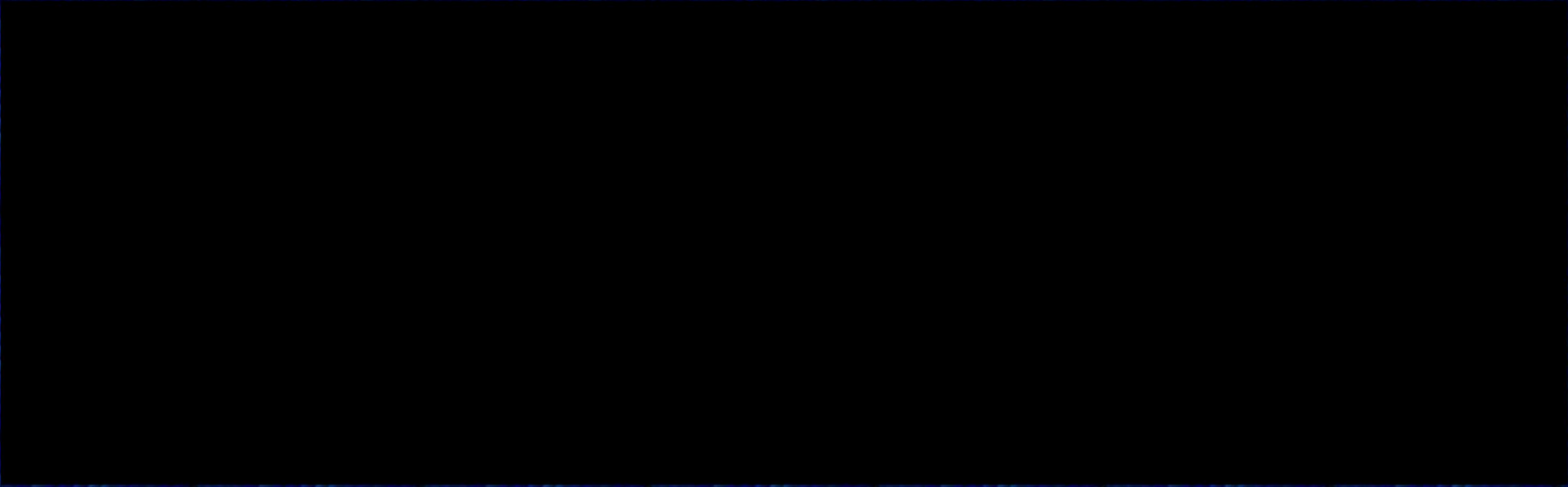
Projections - /Users/ritvik/4_14_projections_hashed/FoFApp.sts

PROJECTIONS

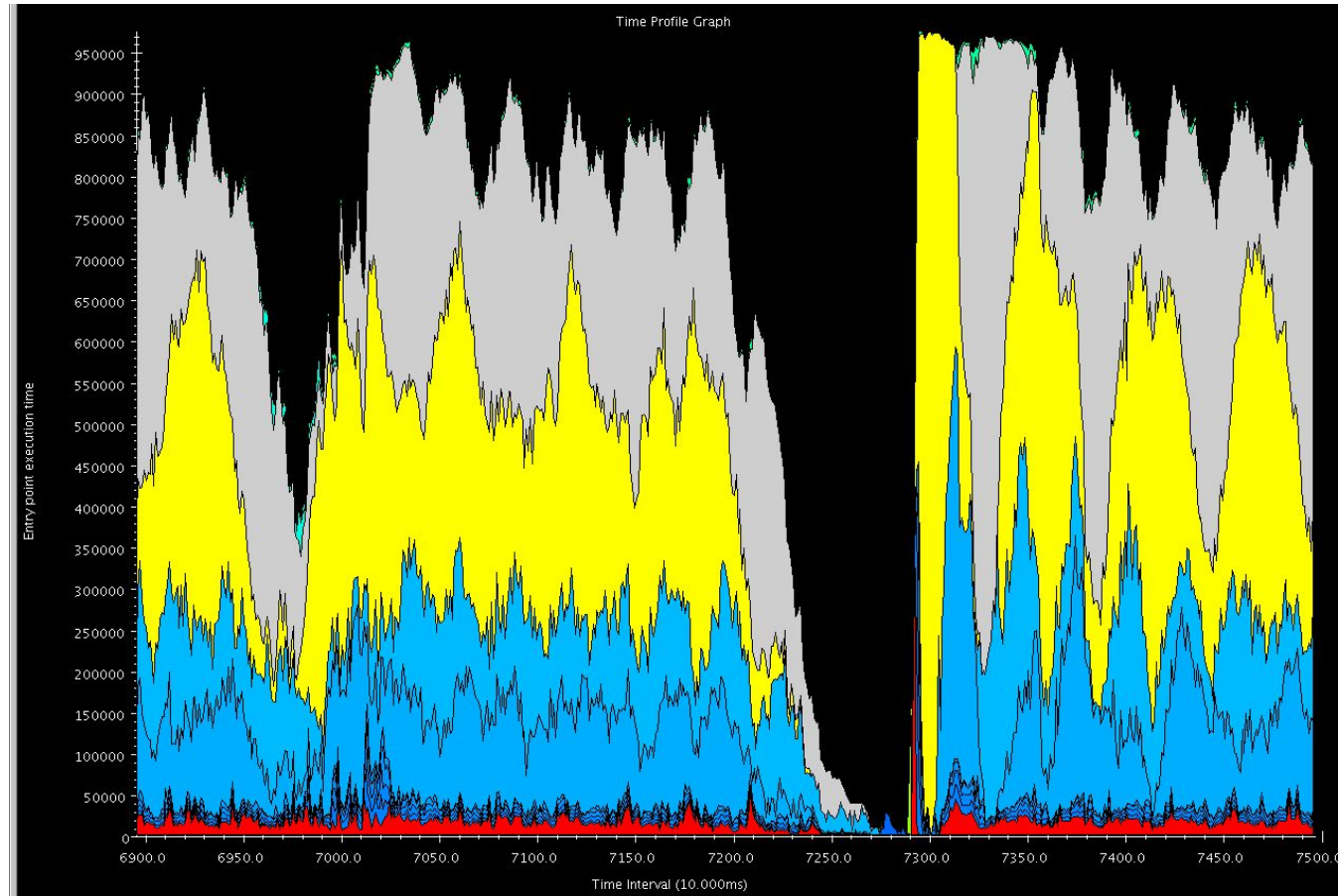
Name: /Users/ritvik/4_14_projections_hashed/FoFApp
Machine layer: ucx-linux-x86_64
Charm++
Version: v8.0.1-devel-33-g1fa944e33
Username: x-rrao
Hostname: a361.anvil.rcac.purdue.edu
Commandline: /home/x-rrao/paratreet/examples/./FoFApp -f /anvil/projects/x-asc050025/x-rrao/globus_shared/lambb.00500 -ll 0.00046 +setcpuaaffinity +traceroot fof_projections +logsize 100000000
Run started at: April 14, 2026, 11:05:48 AM CDT

Active Run: Tab Index 0: Not Applicable

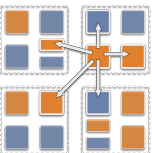
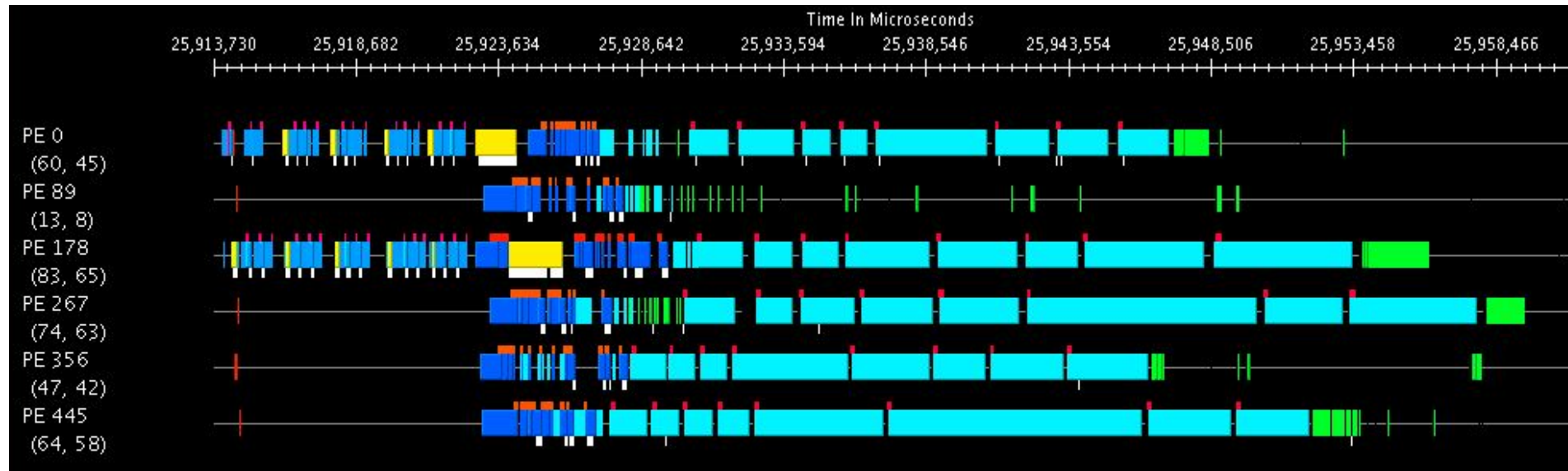
No Summary Data Available



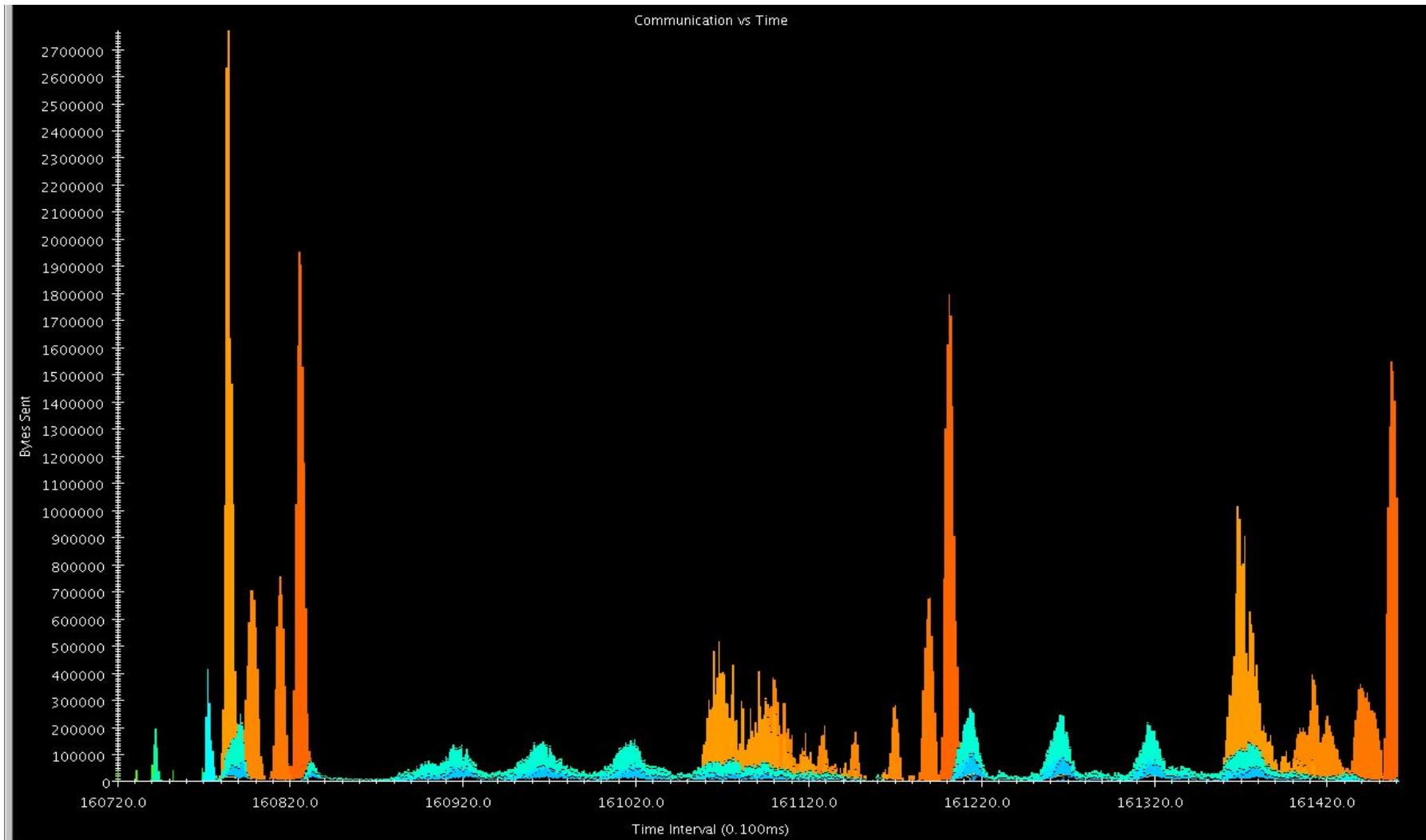
Example: Time profile



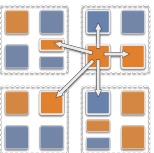
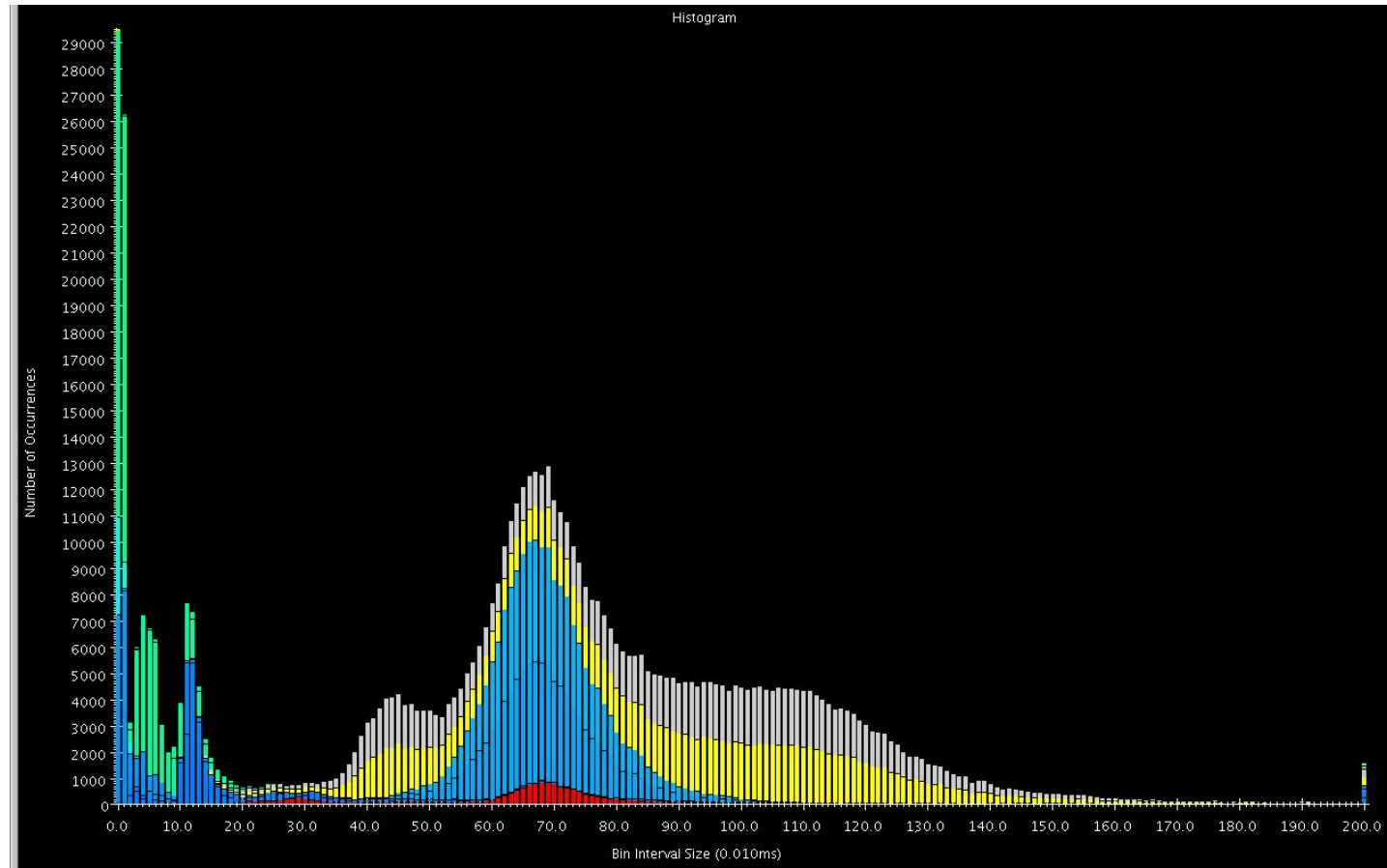
Example: Timeline



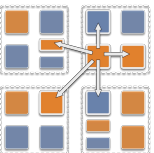
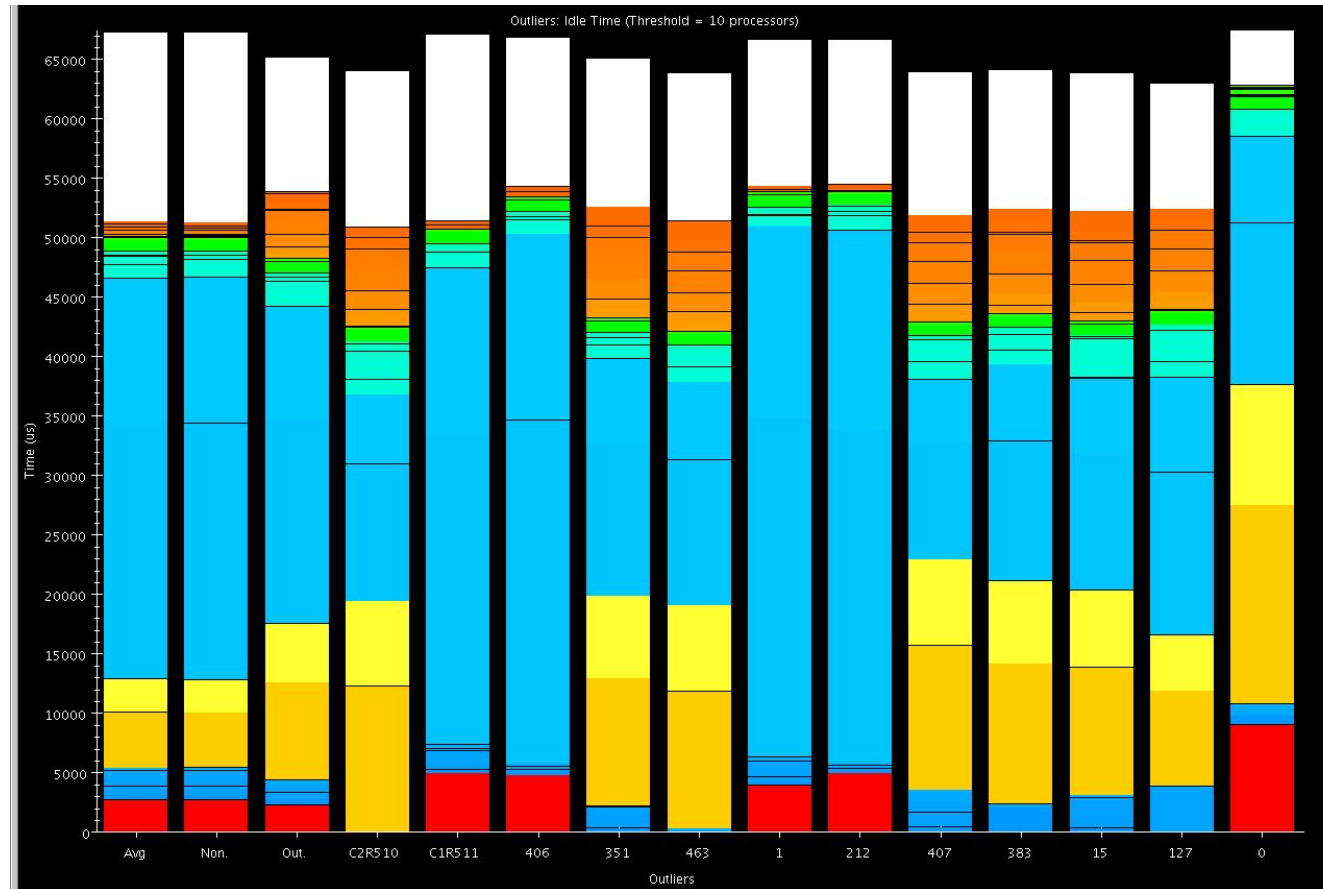
Example: Communication vs. time



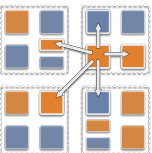
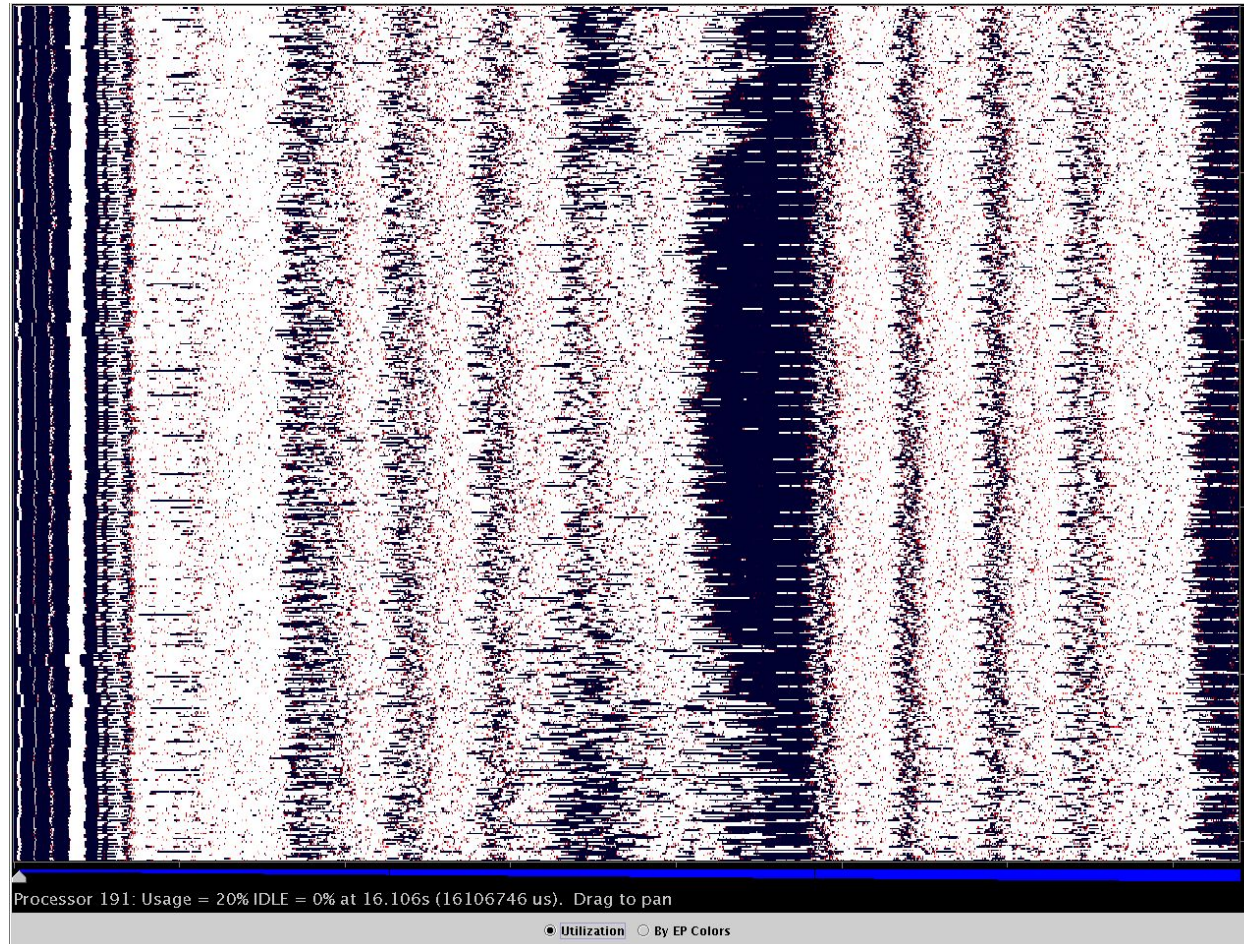
Example: Histograms



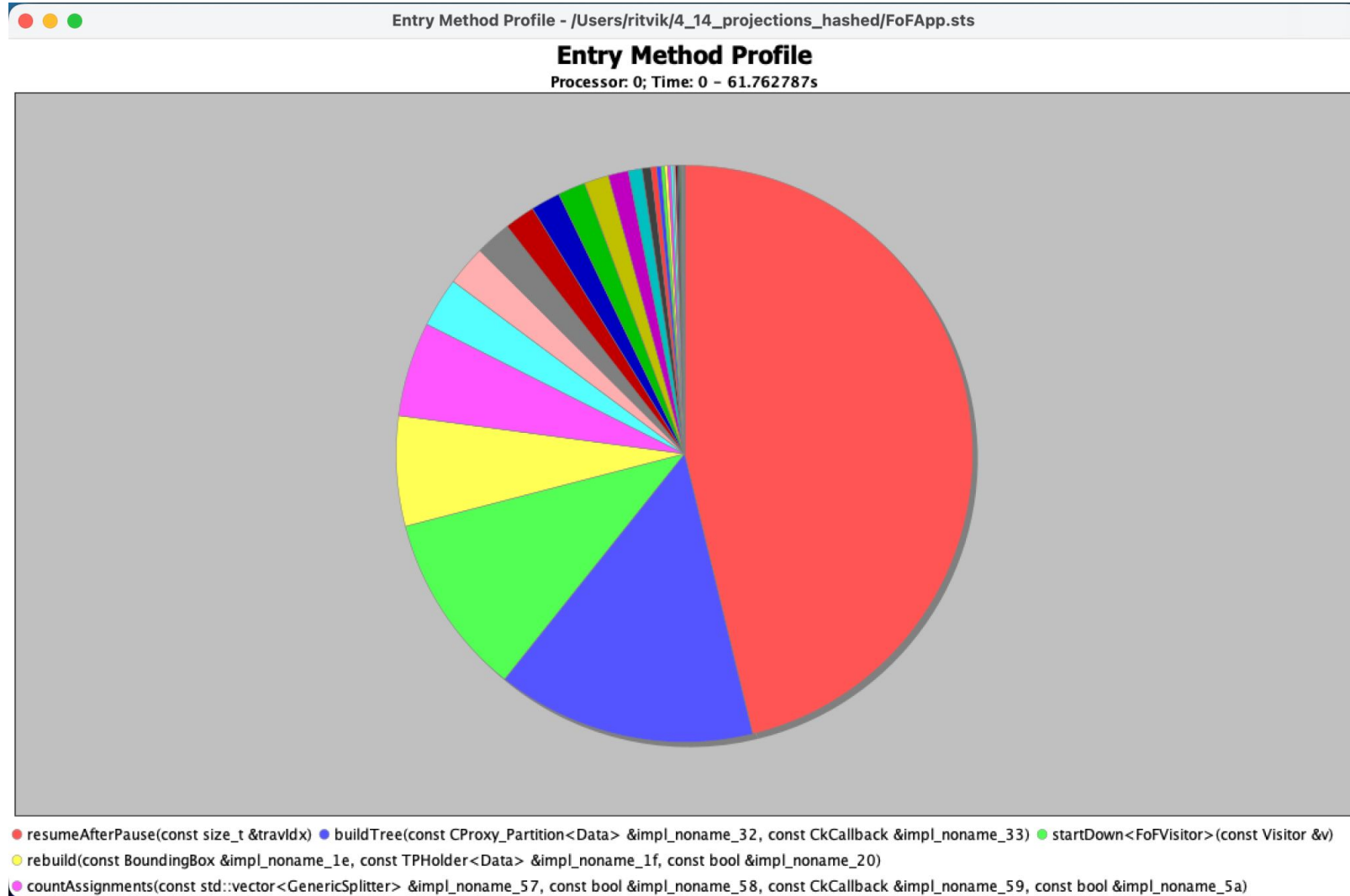
Example: Usage profile



Example: Overview



Example: Entry method profile



Example: Message size evolution

